

PROCESS BOOK · MILESTONE 3

Global Cyber Incident Patterns

An interactive, narrative-driven view of 4,374 documented cyber operations

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Live demo com-480-data-visualization.github.io/visec/src/
Source github.com/com-480-data-visualization/visec
Dataset [EuRepoC Global Dataset of Cyber Incidents](#)

Contents

1	Abstract	1
2	Motivation & Research Questions	1
3	Related Work & Inspiration	1
4	Dataset	1
4.1	Source, scope, and cleaning	1
4.2	Retained columns	2
5	Design Process	2
5.1	Milestone 1 — Exploratory analysis	2
5.2	Milestone 2 — Wireframes and static prototype	3
5.3	Milestone 3 — Interactive narrative	4
5.4	Challenges & how we resolved them	5
6	Visualization Design	5
6.1	Marks, channels, and graphical integrity	5
6.2	Interactions	5
6.3	Timeline annotations	6
6.4	Map design	6
6.5	Storytelling structure	6
7	Implementation	6
7.1	Architecture	6
7.2	Data pipeline & accessibility	6
8	The Final Views	7
9	Evaluation & Lessons Learned	7
10	Team Contributions	7
11	References	8

1 Abstract

Global Cyber Incident Patterns is an interactive, narrative-driven visualization of the EuRepoC dataset of cyber operations, condensing 4,374 incidents across 211 countries (2000–2025) into six coordinated views: an annotated timeline with a brush selector and named real-world event markers, two ranked bar charts (most targeted countries, most frequent incident types — the latter now clickable as a third filter dimension), an attacker–target relationship view, a sectors-and-actors view that exposes target categories and initiator types, and a choropleth world map with arc-based attack flows. A dedicated methodology section explains how the record was assembled and how to read it. This book documents the full design arc from exploratory analysis (M1) through wireframes (M2) to the final interactive site (M3), justifying each design decision in terms of intent and visual encoding.

2 Motivation & Research Questions

Cyber incidents have become central to international relations, yet the global landscape is hard to interpret: data is scattered, attribution is slow, and the EuRepoC export contains 84 attributes per incident. Our goal is to surface the most interpretable structures — who is targeted, who initiates operations, which pairs recur, how activity evolved, how it distributes geographically, and what kinds of actors and sectors sit behind the country-level numbers — for an audience spanning students, researchers, analysts, and policymakers.

The six sections of the site answer six focused questions:

- **Evolution** — How has recorded cyber-incident volume changed from 2000 to 2025, and which named events drove the visible spikes?
- **Targets** — Which countries are targeted most, and how concentrated is that distribution?
- **Incident types** — Which operation categories dominate — and does that match media narratives?
- **Relationships** — Which attacker–target pairs recur, and what does that look like per country?
- **Sectors & actors** — Which target sectors absorb most incidents, and which kinds of actors initiate them?
- **Geography** — Where do incidents flow, and what does the spatial pattern reveal?

3 Related Work & Inspiration

EuRepoC provides a public table view and faceted dashboard optimized for incident lookup (locate, drill down), not for communicating overall structure. Our project takes the opposite approach: we reduced the dataset to 13 columns and committed to a small number of focused, comparable views organized as a narrative.

Design inspirations: **Our World in Data** (clarity, active titles, paired prose); **FT Visual Vocabulary** (chart type follows message, not dataset shape); **Storytelling with Data** (sorted bars, strategic color, de-cluttering); **Tufte** (data-ink ratio, lie factor = 1, proportional channels).

4 Dataset

4.1 Source, scope, and cleaning

The EuRepoC Global Dataset of Cyber Incidents provides 84 columns per row. After filtering to parseable start years in 2000–2025, the working set contains 4,374 incidents across 211 countries. `refactor_data.py` reduces this to 13 analytical columns (committed as `cyber_incidents_ref.csv`, ≈1.9 MB); the site fetches this CSV at runtime with no server-side processing.

4.2 Retained columns

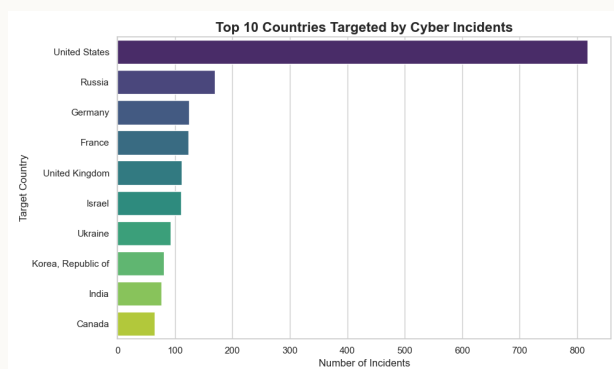
Column	Meaning
start_date	Incident date (year parsed)
incident_type	Operation classification (multi-value, ;-separated)
receiver_country	Target country (multi-value)
receiver_region / category	Geographic region; target type
initiator_country / category	Initiating country and actor type
data_theft, has_disruption, disruption, hijacking	Impact indicators
economic_impact, impact_indicator_value	Economic and quantitative impact

Multi-valued fields are split client-side; country names are normalized via an alias map (e.g. *Russian Federation* → *Russia*). The two *category* columns (*receiver_category*, *initiator_category*) carry verbose EuRepoC labels which we map to short, scannable names (e.g. “*State institutions / political system*” → “*State / political*”, “*Non-state actor, state-affiliation suggested*” → “*State-affiliated*”); the same fields can hold multi-target rows separated by “ - ” which the parser splits on.

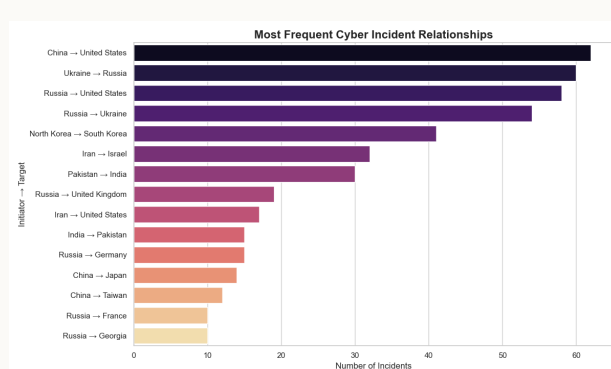
5 Design Process

5.1 Milestone 1 — Exploratory analysis

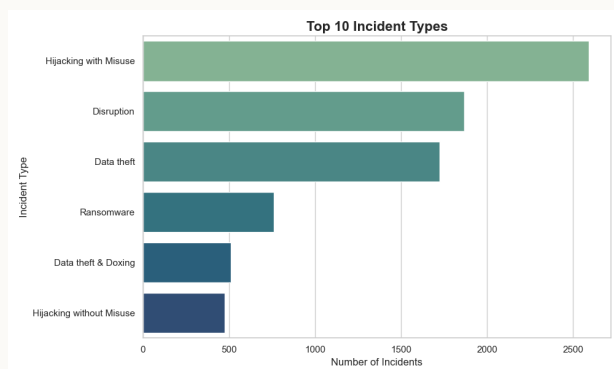
Four exploratory figures verified that the analytical questions have non-trivial answers in the data.



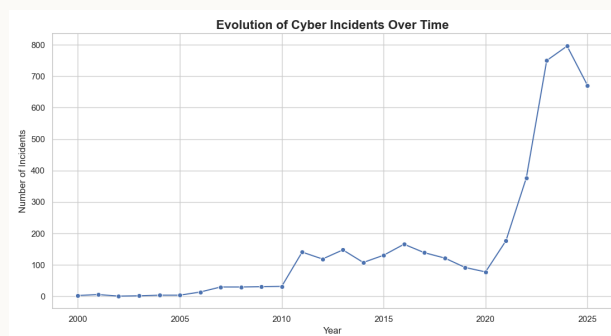
(a) Top targeted countries. US dominates; European and conflict-adjacent countries cluster behind. Motivated Section 2.



(b) Most frequent attacker–target pairs. China/Russia → US lead; Russia → Ukraine follows. Motivated Section 4.



(a) Top incident types. Hijacking, disruption, and data theft dominate; ransomware is under-represented. Shaped Section 3.



(b) Incidents over time. Sharp rises after 2014 and 2020 motivated the timeline brush in Section 1.

5.2 Milestone 2 — Wireframes and static prototype

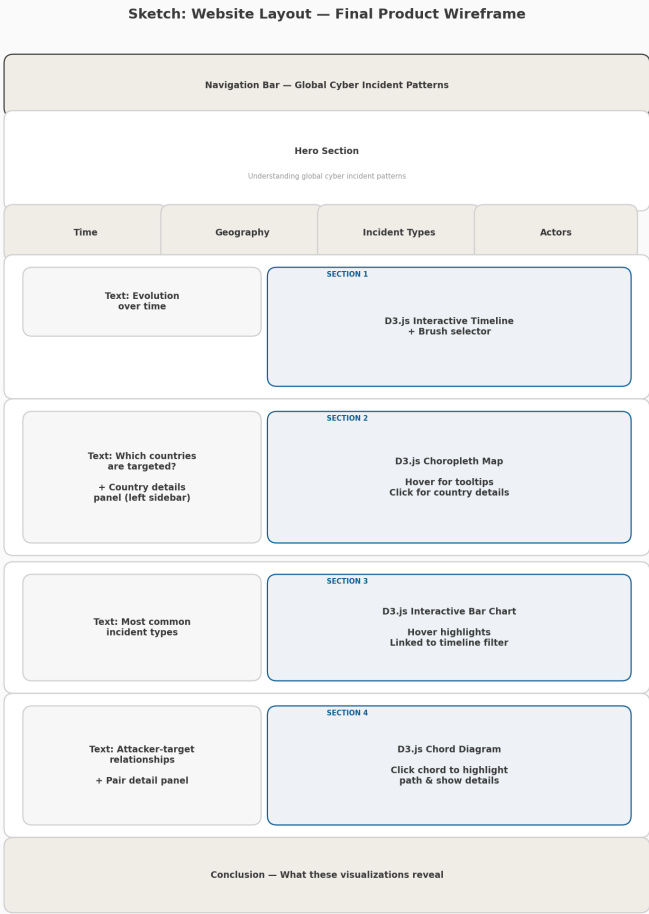
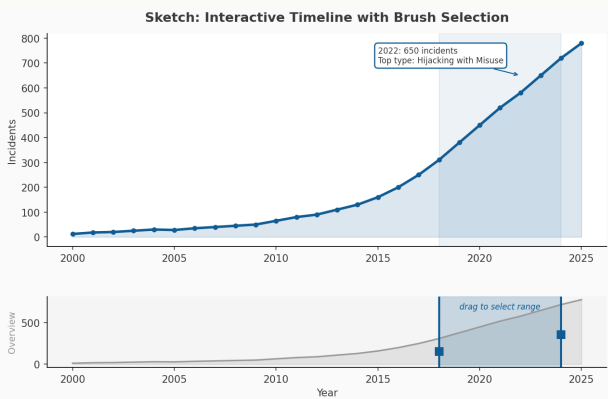
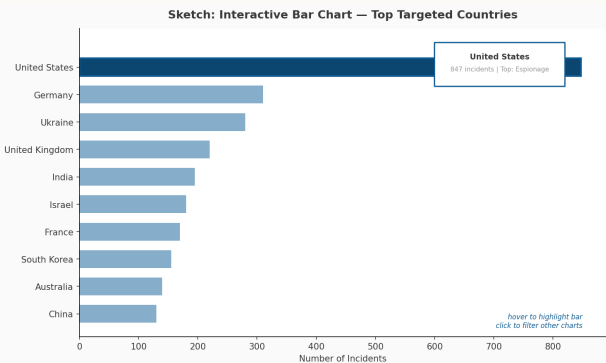


Figure 3 — Overall layout: hero, analytical sections with paired prose and charts, sticky nav. Map drafted as the optional final section.

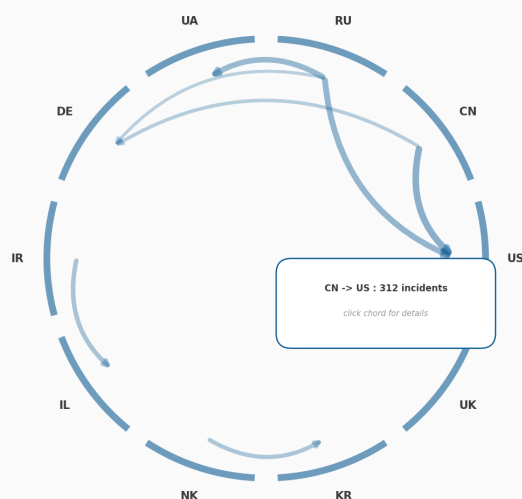


(a) Timeline sketch: a simple line chart with a draggable brush below it. In the final site this became a filled area+line chart, gained seven named event annotations rendered as purple tags above the line, and the brush was extended with precise year-range labels. The core mechanic — brush drives all downstream views — was kept exactly as sketched.



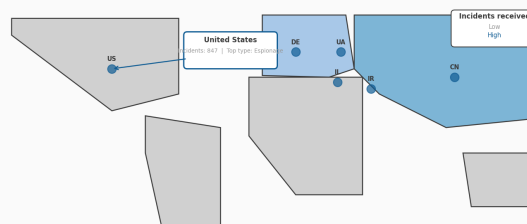
(b) Horizontal bar chart sketch: sorted by value, text labels left-aligned, single accent color. The final implementation is faithful to this idiom and intentionally re-uses it for all five ranking views (targets, types, pairs, sectors, actors) so the reader never has to re-learn the encoding. The only addition was a clickable interaction on the incident-type bars to act as a third cross-chart filter.

Sketch: Chord Diagram — Attacker-Target Flows



(a) Chord diagram, prototyped in M2 for the attacker–target relationship view. After building a working version we found that arc width — the diagram’s primary channel — is harder to compare accurately than bar length, and the layout required substantial cognitive setup for non-specialist audiences. We replaced it with paired horizontal bar panels (who targets this country; who this country targets), which re-use the encoding the reader already knows from Section 2.

Sketch: Interactive Choropleth World Map



(b) World map sketch: a plain choropleth with no directional information. The final implementation added arc-based attack-flow arrows (red incoming, orange outgoing) that appear on hover, because the sketch made clear that a pure luminance encoding cannot convey direction. The Natural Earth projection replaced the initially considered Mercator to avoid area distortion at high latitudes.

5.3 Milestone 3 — Interactive narrative

M3 turned the static skeleton into a fully coordinated interactive site. The upgrades fall into five buckets:

Three-axis coordination. Timeline brush, country click/search, and (new) incident-type click each drive every downstream view through a shared state object. Each chart’s header carries a filter chip that summarises the active state and lets the reader remove filters individually; a *Reset all* button clears everything in one click.

Annotated timeline. Seven named historical events — Stuxnet (2010), Sony Pictures (2014), WannaCry / NotPetya (2017), SolarWinds (2020), Colonial Pipeline (2021), Russia–Ukraine cyber front (2022), MOVEit (2023) — appear as small purple tags above the line with hover tooltips. The spikes the chart already showed now have stories attached.

New view: sectors & actors. A dedicated section pairs two ranked bar charts built from previously-unused columns: *receiver_category* (target sector) and *initiator_category* (attacker actor type). State institutions and critical infrastructure dominate the target side; non-state and state-affiliated groups dominate the attacker side.

World map. Choropleth plus arc-flow overlay (red incoming, orange outgoing). Inherits every upstream filter — brushing the timeline or selecting an incident type immediately recolors the map.

Methodology & editorial polish. A dedicated *How this record is made* section explains the dataset’s construction and reading rules. Section titles were rewritten from descriptive to active, and a country-focus card, an incident-type breakdown card, and a dynamic key-takeaway block were added.

5.4 Challenges & how we resolved them

Multi-valued fields and row explosion. Many EuRepoC rows list several countries or incident types separated by semicolons. Naively counting rows would systematically under-count incidents affecting multiple targets. We resolved this by splitting every multi-valued cell client-side in `buildIncident()` and treating each atomic value as an independent data point, accepting that aggregate counts can exceed the raw row count. This decision is explained in the methodology section so readers can calibrate the numbers.

Country-name normalization across 211 territories. The dataset, the D3 world-atlas TopoJSON, and ISO-3166 all use different name conventions. A single missing alias left a country unjoined and invisible on the map. We maintained a hand-curated `NAME_TO_ISO` map covering ~150 aliases; every map regression was traced to a missing entry. The lesson: build a validation script that logs unmatched names on load rather than silently dropping them.

Chord diagram vs. bar panels for relationships. A working chord diagram was built in M2 but abandoned after walkthroughs showed that non-specialist readers spent more time decoding the layout than reading the data. The replacement — two ranked horizontal bar panels conditioned on the selected country — re-used the existing bar-chart component and required no new visual vocabulary.

Cross-chart filter composition. Supporting three simultaneous filter axes (year range, country, incident type) that compose correctly required all aggregation functions to read from a single state object and be called through one `renderAll()` entry point. The architecture was established early in M3 and made adding the incident-type axis a matter of adding one field to state and one branch in each aggregator — a two-hour addition rather than a rewrite.

Performance at interaction time. Recomputing all aggregates on every brush event over ~4,400 rows caused noticeable jank on slow devices. We debounced the brush handler to 40 ms and moved the heaviest aggregation (`buildRelationshipData`) behind a `requestAnimationFrame` call, restoring smooth 60 fps transitions without sacrificing correctness.

6 Visualization Design

6.1 Marks, channels, and graphical integrity

All ranking views use **horizontal bar charts**: position on a common scale and length are the most accurately perceived quantitative channels, and we apply the same idiom across targets, incident types, attacker–target pairs, target sectors, and initiator actors so the reader can compare across charts without re-learning the encoding. **Hue** encodes direction on the map (red incoming, orange outgoing) and discriminates the per-country relationship panels. **Luminance** (sequential blue) encodes incident count on the choropleth. Area is avoided except in tooltips, where it is always paired with a numeric label.

Integrity rules applied throughout: axes start at zero (lie factor = 1); bars are sorted by value; gridlines are removed; active titles state the analytical point of each chart (e.g. “What dominates the record is not what dominates the headlines”).

6.2 Interactions

The site follows the *overview → zoom & filter → details on demand* mantra. A single state object drives all views and now carries three filter dimensions plus a per-relationship selection:

```
const state = { yearRange: [2000,2025], selectedCountry: null,
  selectedRelationship: null, selectedIncidentType: null };
```

Any interaction mutates this state and calls `renderAll()`, making cross-chart coordination trivial. The three filter axes compose: brushing 2017–2022 then clicking *Ransomware* in Section 3 then clicking *Ukraine* in Section 2 narrows every chart, the map, and the takeaway texts to that exact slice. Transitions are 200–400 ms with a single easing. The chord diagram prototyped in M2 was dropped: arc width is a weaker channel than bar length for the ranking task, and the diagram requires significant cognitive setup for non-specialist audiences.

6.3 Timeline annotations

The timeline carries small purple event tags anchored on the year axis, with dashed vertical guides running down to the X axis. Hovering a tag reveals a tooltip naming the event (e.g. *“MOVEit mass exploit”*) and explaining its role (e.g. *“A single vulnerability used to compromise thousands of organisations; large boost to recorded data-theft incidents”*). The annotation layer respects the active year window, so brushing a sub-period only renders the tags inside it.

6.4 Map design

Natural Earth projection (Mercator avoided for area distortion at high latitudes). World atlas loaded as 110 m TopoJSON (smaller than equivalent GeoJSON). A NAME_TO_ISO mapping for ~150 countries bridges dataset names to world-atlas ISO codes. Arc arrows add directional information that a pure choropleth cannot carry.

6.5 Storytelling structure

Six numbered sections read top-to-bottom, followed by a methodology section and a closing summary. Each analytical section pairs explanatory prose with an interactive chart and a key-takeaway block. Numbers are rendered for legibility (“1,400+” not “1,427”). The closing section explicitly states what the data shows *and* what it cannot show, and the methodology section explains the recording process so the reader can calibrate every chart against the dataset’s limits.

7 Implementation

7.1 Architecture

Fully static, no-build-step site: `index.html` , `script.js` , `style.css` . All dependencies loaded from CDN.

Library	Version	Purpose
D3.js	v7	Scales, axes, brush, geo, all charts
topojson-client	v3	Decoding world-atlas TopoJSON
world-atlas	v2	110 m country geometry

Deployment is a single `git push`; GitHub Pages serves `src/` directly.

7.2 Data pipeline & accessibility

`d3.csv()` fetches the cleaned CSV on load; `buildIncident()` parses each row into a typed object including the new `targetSectors` and `initiatorActors` arrays (built by `splitCategories()`, which splits on `;` and the `multi-target` – separator and applies a label-shortening map). Aggregates (`buildTargetData`, `buildTypeData`, `buildSectorData`, `buildActorData`) are recomputed on every interaction (safe at ~4,400 rows). Accessibility: `aria-live="polite"` region updated on filter changes; filter chips are real `<button>` elements; SVGs rebuild (not scale) on resize.

8

The Final Views

- Section 1 — Evolution.** Area+line chart, 2000–2025, with brush selector. Hovering any year shows the exact count. Seven named historical events appear as purple tags above the line, with hover tooltips. Title: “More incidents, but not the full picture.”
- Section 2 — Targets.** Top-10 targeted countries, ranked horizontal bars. Bar click or search input selects a country and propagates through all downstream views.
- Section 3 — Incident types.** Most frequent incident types (multi-value cells split into atomic labels). **Bars are now clickable:** selecting a type filters every other view to incidents of that type. Click again to clear.
- Section 4 — Relationships.** Default: top-12 attacker–target pairs. With a country selected: splits into *who targets it* and *who it targets* panels.
- Section 5 — Sectors & actors.** Two stacked bar charts using previously-unused fields. *Target sectors* (top: state / political, critical infrastructure, corporate) sits next to *initiator actor types* (top: non-state group, state-affiliated, state). A live narrative card describes the current slice and respects every active filter.
- Section 6 — World map.** Choropleth by incident count + arc arrows on hover (red = top attacker in, orange = top target out). Respects all upstream filters.
- Method — How this record is made.** Six methodology cards covering source (EuRepoC), what is in the dataset, what is not, attribution levels, multi-valued fields, and how to read the trends. Designed to pre-empt the questions a careful reader will have at the end of the scroll.

9

Evaluation & Lessons Learned

- What worked.** The single state object made cross-chart coordination almost trivial — adding a third filter axis (incident type) was a few lines because every renderer already reads from one source of truth. Active editorial titles measurably improved perceived clarity in walkthroughs. Building the map last (on top of an already-working state model) gave it brush-and-country filtering for free. Re-using the bar-chart component for the new Section 5 (sectors and actors) meant zero new visual vocabulary for the reader to learn. The no-build-step architecture saved hours of tooling overhead.
- Limitations.** The dataset records only attributed and disclosed incidents. State-sponsored operations that go undetected, undisclosed breaches, and contested attributions are absent — a point the methodology section now states explicitly. Maintaining a hand-curated NAME_TO_ISO map was the most fragile part of the project; a single missing alias produced an unjoined country on the map.

10

Team Contributions

Member	Primary ownership
Ziyad Baghoury	Data preparation and cleaning pipeline (<code>refactor_data.py</code>); exploratory analysis figures (top targets, attack pairs, incident types, time evolution); world map end-to-end: choropleth encoding, Natural Earth projection, arc-arrow attack-flow overlay, TopoJSON integration, NAME_TO_ISO country normalization; country search bar; narrative texts across all six analytical sections.
Rayane Bentahar	Incident-type frequency and time-evolution exploratory plots; related work section; static prototype of the full website layout; timeline brush (period selector); cross-chart country coordination — clicking a country bar propagates the selection across all views; attacker/receiver panels in the relationships section; incident-type click filter — selecting a type updates the relationship panels to reflect attacks sent and received under that category.
Ayman Bakiri	Timeline area+line chart, brush selector, and event annotations (named tags, hover tooltips, dashed guides); accessibility audit (ARIA labels, keyboard navigation, responsive SVG rebuild on resize); GitHub Pages deployment and static site architecture; video presentation.
Mohamed Sami Ghrab	Design system (color palette, typography, layout tokens); sectors & actors charts and narrative; methodology cards; cross-browser compatibility testing and performance profiling (brush debounce, <code>requestAnimationFrame</code> optimisation); process book.

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